

Understanding the periodic table

THERE ARE TRENDS IN THE PERIODIC TABLE.

The periodic table arranges the elements according to the arrangement of their atoms' electrons. This means that similar elements are grouped together.

Size of atoms

Atoms get bigger as you move down the table because each period, or row, begins when a new shell is added to the atom. However, they get smaller from left to right as the number of outer electrons increases. This is because atoms with more outer electrons are held together with greater force, pulling them into smaller volumes.

Li
LITHIUM

Two shells
A lithium atom has three electrons, with the third located in the second shell.

Na
SODIUM

Three shells
The outer electron of the sodium atom starts a third shell, making the atom larger.

K
POTASSIUM

Four shells
Potassium, in Period 4, has a fourth electron shell located far out from the nucleus.

The diagram illustrates the periodic trend of increasing atomic size. It features three vertical representations of atoms: Lithium (Li) at the top, Sodium (Na) in the middle, and Potassium (K) at the bottom. Each atom is shown with its nucleus (a red circle with a plus sign) and concentric green circles representing electron shells. Lithium has two shells and three electrons (two in the first shell, one in the second). Sodium has three shells and eleven electrons (two in the first, eight in the second, one in the third). Potassium has four shells and nineteen electrons (two in the first, eight in the second, eight in the third, one in the fourth). Arrows point from the text descriptions to the corresponding atom diagrams. The overall size of the atoms increases from top to bottom, with Potassium being the largest and Lithium the smallest.

The diagram illustrates the atomic structure of three elements: Sodium (Na), Magnesium (Mg), and Aluminium (Al). Each atom is shown with its nucleus and three concentric electron shells. The outermost shell contains 1, 2, and 3 electrons respectively for Na, Mg, and Al. Arrows point from the text descriptions below to the outer shells of the atoms.

Element	Outer Electrons	Effect on Atomic Size
Sodium (Na)	1	one outer electron does not pull the atom very tightly
Magnesium (Mg)	2	two outer electrons pull the sphere smaller
Aluminium (Al)	3	three outer electrons pull the sphere even smaller

SEE ALSO	
⟨ 113 Reactivity	
⟨ 116–117 Periodic table	
Inside atoms	168–169 ⟩

1																
1	H HYDROGEN															
2	Li LITHIUM	Be BERYLLIUM														
3	Na SODIUM	Mg MAGNESIUM														
4	K POTASSIUM	Ca CALCIUM	Sc SCANDIUM	Ti TITANIUM	V VANADIUM	Cr CHROMIUM	Mn MANGANESE	Fe IRON								
5	Rb RUBIDIUM	Sr STRONTIUM	Y YTTRIUM	Zr ZIRCONIUM	Nb NIOBIUM	Mo MOLYBDENUM	Tc TECHNETIUM	Ru RUTHENIUM								
6	Cs CESIUM	Ba BARIUM	La-Lu LANTHANIDE	Hf HAFNIUM	Ta TANTALUM	W TUNGSTEN	Re RHENIUM	Os OSMIUM								
7	Fr FRANCIUM	Ra RADIUM	Ac-Lr ACTINIDE	Rf RUTHERFORDIUM	Db DUBNIUM	Sg SEABORGIUM	Bh BOHRVIUM	Hs HASSIUM								
			La LANTHANUM	Ce CERIUM	Pr PRASEODYMIUM	Nd NEODYMIUM	Pm PROMETHIUM									
			Ac ACTINIUM	Th THORIUM	Pa PROTACTINIUM	U URANIUM	Np NEPTUNIUM									

Metals and nonmetals

The left side of the periodic table is made up of metallic elements; the right side, nonmetallic. A metallic element has atoms that give up their outer electrons easily. The nonmetals hold firmly to their outer electrons and have very different properties from metals. Eight elements are semimetals, which have characteristics of both metals and nonmetals.

METALLIC AND NONMETALLIC	
Metallic	Nonmetallic
conducts heat	good insulator
conducts electricity	resists current
malleable and tough	brittle and crumbly
shiny and opaque	dull and translucent
high density	low density
low ionization energy	high ionization energy

◀ **Metallic vs nonmetallic**

Metal atoms have a certain set of characteristics due to their atomic structure. Nonmetals have an almost opposite set of characteristics.